

IoT-Based Smart Kitchen Safety and Fan Automation System

Arduino Pin Connections

Component	Arduino Pin(s)	Power	Notes
MQ-2 Gas Sensor	A0	5V, GND	Analog gas level output
DHT11 Sensor	D2	5V, GND	Temperature & humidity data
16x2 LCD (I2C)	SDA→A4, SCL→A5	5V, GND	I2C interface
Relay Module	D8	5V, GND	Controls DC fan
Buzzer	D9	5V/GND	Alarm output
LED	D10	220Ω resistor to GND	Status indicator
DC Fan	Relay NO/COM	External 5V	Switched by relay
Arduino Uno	USB/9V DC Jack	—	Main controller